Colour-blind Series: Deuteranopia

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What is it for?

Deuteranopia is a form of colour blindness that affects the ability to see green. Like Protanopia, it's a type of 'red-green colour blindness'. Protanopia affects the cells that sense red. Deuteranopia affects the cells that sense green. It's the more common of the two, affecting about 1% of men. It's very rare in women.

In practice, the experience is similar. Red and green become very difficult to tell apart, making colour-coded information unreliable. A traffic light style status indicator, a heatmap, or a form validation message relying on red and green will all present the same challenge.

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)